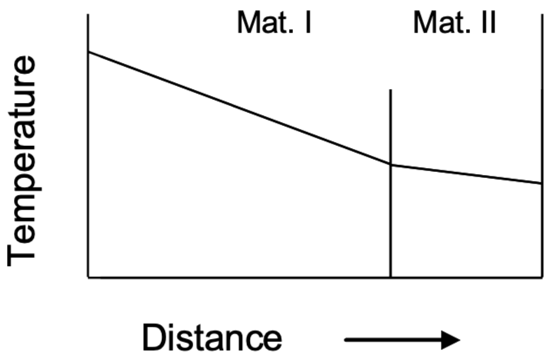
CHE 311H HW #6
Fall 2023

Instructions: Show all work and box in your final answer. Write all group member's names on the first page and the group number in top corner of all subsequent pages. Remember, illegible solutions will receive no credit. Additionally, regrade requests must be submitted within 2 weeks of the due date.

Due date: Thursday, November 2, 2023, during lecture. Submit as a hardcopy.

1. BSL 10A.6

2. [Old Exam problem] At steady state, the temperature profile through a laminate of two slabs is shown below. The temperature drop (or difference) across Material I is 10 times that across Material II. However, Material I is twice as thick as Material II. How much smaller or larger is the *thermal conductivity* of Material I relative to Material II?



3. [Old Exam problem] Consider radial heat flow out of a composite cylinder shown below. The heat flux out of the outer cylinder, q_3, is known, and so are the radii of the different layers. Your friend from Georgia Tech is struggling to determine the heat fluxes q_2 and q_1 in terms of known quantities. This is a trivial problem for you all. Please enlighten him with your answer.